Berlin Heart®



EXCOR® Pediatric

Anticoagulation

Disclaimer



This slide set is intended to provide a detailed overview on the use of the Berlin Heart EXCOR® VAD system and its components.

This presentation does not relieve the clinician of their responsibility to read and understand the Instructions for Use (IFU) provided with each system.

It is also the responsibility of the clinician to properly train the patient/parent on the safety risks and cautionary measures described in the IFU.

Agenda

- Overview anticoagulation for EXCOR® patients
- Edmonton protocol
- Alternative anticoagulation protocols
- Special cases

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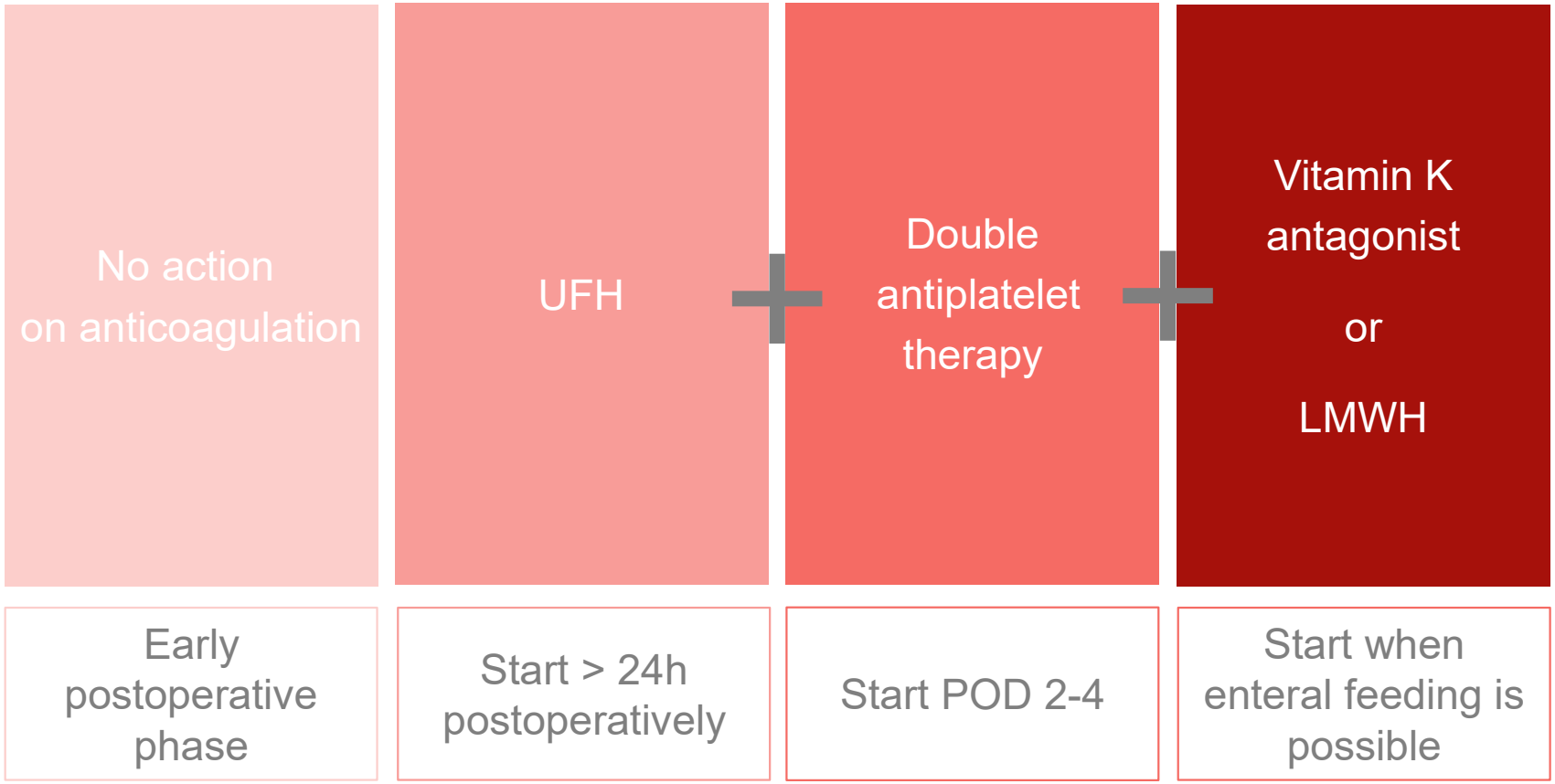
Pre-operative examinations

- Genetic factors of coagulation abnormalities
  - von Willebrand disease
  - Factor V Leiden mutation
  - Protein C/S – deficiency
  - decreased levels of natural anticoagulants (e.g. antithrombin III)

Intra-operative / before leaving OR

Completely reverse anticoagulation !

EXCOR® Pediatric – Overview anticoagulation



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EXCOR® Pediatric - UFH

- **Start** 24 - 48 hrs post-OP (no bleeding or at least less than 5 ml/h). Application should be delayed if ECLS was used (about 48 hrs postoperatively).
- **Lab results:** Hb, Hct, Plt > 20.000/µl, ATIII-Level > 70% etc.
- **TEG analysis:** before start if possible → normal coagulation
- **aPTT check**: every 6 hours

	< 1 year	> 1 year
Initial dosage i.v. **(no bolus)**	15 U/kg/h	10 U/kg/h
Maintain (after PTT control every 6 hr)	↑ 28 U/kg/h max.	↑ 20 U/kg/h max.

Target

PTT 60-85 sec. or Anti-FXa-Level 0.35 – 0.5 U/ml

EXCOR® Pediatric - Antiaggregation

- Platelet count min. > 40 000/µl
- Dipyridamol on POD 2 → 4 mg/kg/d (divided in 4 doses), max.15 mg/kg/d
- ASA on POD 4 (if drainages are out) → 1 mg/kg (devided in 2 doses) if aggregation test available otherwise standard 4-5 mg/kg/d

Target

ARA activity < 30%

EXCOR® Pediatric - Vitamin K Antagonist

- Start VKA (e.g. Warfarin) when enteral feeding possible + age > 12 months
- Initial dosage → 0.2 mg/kg/d (max. 5 mg/d) / INR (Daily control first 4 weeks, then 1-2x/week)

INR	Warfarin Adjustment
1.1-1.3	Repeat initial loading dose
1.4-1.9	50% of initial loading dose
2.0-3.0	50% of initial loading dose
3.1-3.5	25% of initial loading dose
>3.5	Skip dose until INR <3.5

dose of Warfarin for maintaining a target INR 2.7 – 3.5

EXCOR® Pediatric - LMWH

Target INR 2.7 – 3.5

- If INR is out of target range under VKA treatment

	INR < 2.7 - > 2.0 (prophylactic)	INR < 2.0 (therapeutic)
e.g. Enoxaparin	0.5 mg/kg 2x/d	1.0 mg/kg/bw 2x/d

EXCOR® Pediatric - LMWH

- Indications: switch from UFH or if INR is out of target range
- Normal renal function
- Daily control of anti-FXa Level → stable therapeutic level → twice/once weekly

	< 3 month	> 3 month
Initial dosage s.c. (Enoxaparin)	1.8 mg/kg 2x/d	1.1 – 1.4 mg/kg 2x/d
Anti-FXa control 4 hrs after last application		

Target Anti-FXa-Level 0.6 – 1.0 U/ml

Examinations

- Thrombelastography (TEG® or ROTEM®)
  - helpful in the early postoperative period to distinguish between surgical bleeding and diffuse bleeding due to coagulopathy

- PlateletMapping™ or Multiplate™ impedance aggregometry
  - aims to determine to what degree platelet function may be inhibited due to pharmacologic inhibition of either the arachidonic acid (AA) or adenosine diphosphate (ADP) pathways
  - helpful to monitor and adjust antiplatelet therapy

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Stanford Protocol



The Journal of
Heart and Lung
Transplantation
http://www.jhltonline.org

ORIGINAL CLINICAL SCIENCE

Impact of a modified anti-thrombotic guideline on stroke in children supported with a pediatric ventricular assist device

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Stanford Protocol

- Modified anti-thrombotic guideline
  - More intense platelet inhibition
  - Less reliance on platelet function testing

- Aim:
  - To reduce stroke rate in EXCOR® recipients without a simultaneous increase in bleeding
  - To improve the safety and survival of children with advanced heart failure supported with the Berlin Heart EXCOR® pediatric VAD

Stanford Protocol

Table 1 Comparison of Anti-thrombotic Therapy Targets in the Edmonton and Stanford Anti-thrombotic Guidelines for the Berlin Heart EXCOR Pediatric VAD

	Edmonton Guideline	Stanford Guideline
Anti-coagulant	Enoxaparin (anti-Xa 0.6 to 1.0)	Same
Aspirin	Titrated to AA inhibition of >70% by TEG/PM	Titrated to a weight-based dose of 30 mg/kg/day (maximal dose 2,000 mg/day)
Dipyridamole (Persan-tine)	Titrated to an ADP Net G[a] 4 to 8 or ADP inhibition >70% by TEG/PM	Titrated to a weight-based dose of 15 mg/kg/day
Clopidogrel (Plavix)	No recommendation	0.2 mg/kg/day (starting) titrated to a fixed dose of 1 mg/kg/dose once daily
Prednisone	No recommendation	As needed for fibrinogen >600 mg/dl or other signs of inflammation (fever, rise in CRP)[b]

AA, arachidonic acid; ADP, adenosine diphosphate; C-reactive protein; IDE, Investigational Device Exemption (study); INR, international normalized ratio; PM, PlateletMapping; TEG, Thromboelastography; VAD, ventricular assist device.

[a]Net G is calculated by subtracting the percent inhibition of ADP from 100%, dividing by 100, and multiplying the value by the baseline G from the citrated specimen activated with kaolin in the presence of heparinase.

[b]Anti-platelet therapy and steroids titrated primarily to achieve an M_A value of between 55 and 65 mm using a citrated specimen activated with kaolin in the presence of heparinase.

Stanford Protocol - Results

Table 3 Serious Adverse Event Rates by Anti-thrombotic Guideline[a]

	Edmonton Guideline ($N = 16$)	Stanford Guideline ($N = 11$)	Incidence rate ratio (SG:EG)	p-value
Ischemic stroke	6 (4.9)	1 (0.8)	0.16 (0.03 to 1.0)	0.031
Hemorrhagic stroke	0	0	NA	NA
Systemic thromboembolism	1 (0.8)	0	—	—
Bleeding	23 (18.8)	11 (8.6)	0.46 (0.23 to 0.92)	0.015
Pump exchanges	41 (33.6)	21 (16.4)	0.49 (0.29 to 0.82)	0.003
All serious adverse events	71 (58.1)	34 (30.5)	0.52 (0.36 to 0.77)	<0.011

IDE, Investigational Device Exemption trial; NA, not applicable.

[a]Data are presented as the number of events followed by the incidence rate per 1,000 days of support in parentheses, unless otherwise specified. Incidence rate ratios are SG:EG, with 95% confidence intervals in parentheses. Serious adverse events are defined according to the original pediatric INTERMACS definitions. Data are based on 1,220 days of support in the SG cohort and 1,279 in the SG cohort.

Agenda

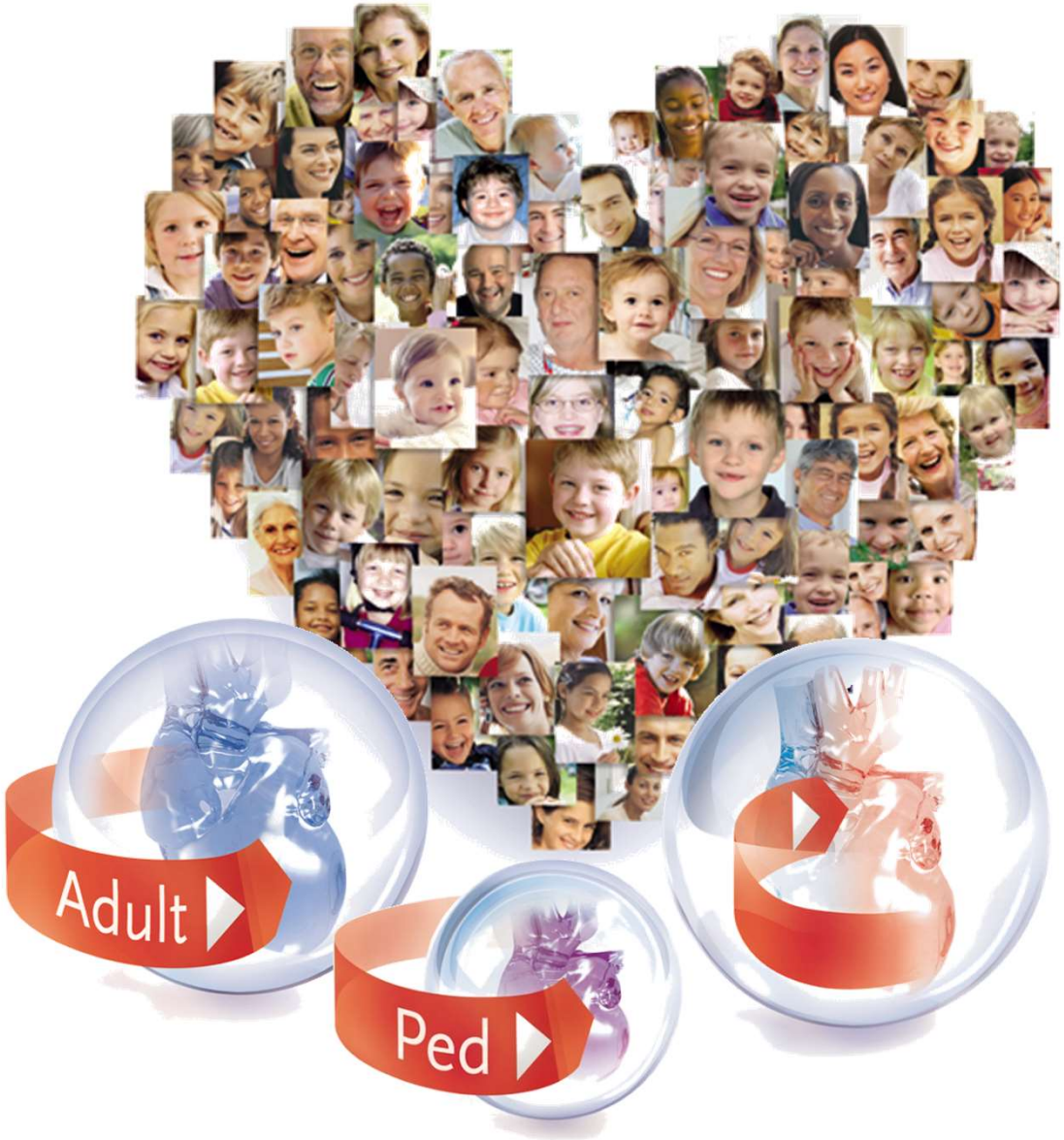
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Heparin Induced Thrombocytopenia (HIT)

- Caused by PF4/Heparin antibodies
- Difficult to distinguish HIT from non-HIT thrombocytopenia after VAD implantation
- Thrombocytopenia: occurs frequently after VAD-implantation
- Reasons: after CPB/cardiac surgery, circulatory failure, infection
- HIT = clinical events + antibodies in temporal relation

Heparin Induced Thrombocytopenia (HIT)

- Alternative anticoagulants
  - **Bivalirudin** (enzymatic metabolization, largely independent from organ function)
  - **Lepirudin** (Refludan, renal metabolization)
  - **Argatroban** (hepatic metabolization)



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